



Questions

These questions use the following database table, PRODUCTS, about products for sale in a shop.

ID number	Name	Type	Cost	Quantity in stock
12GH	Saturn	Chocolate	0.56	100
59RF	Vanilla	Fudge	1.99	20
3WR	Peanuts	Nuts	2.56	2
52D	Jazz	Fruit	0.23	33
6SE	Signet	Fudge	1.48	5

- 5 Write the results returned by the following SQL script:

```
SELECT ID number
FROM PRODUCTS
```

- 6 Write the results returned by the following SQL script:

```
SELECT Name, Type
FROM PRODUCTS
```

- 7 Complete the SQL script to display the prices of all the products.

```
SELECT .....
FROM PRODUCTS
```

- 8 Complete the SQL script to display the ID number and name of all the products.

```
..... ID Number, .....
FROM PRODUCTS
```

Questions

These questions use the following database table, MARKS, about students and their marks in tests.

Student ID	Subject	Test name	Percentage	Grade
123	Science	Physics 1	10	Fail
596	Computer Science	Programming 2	53.3	Pass
123	Maths	Trigonometry 6	10	Pass
023	Maths	Trigonometry 6	50	Pass
802	Science	Physics 1	25	Fail
806	Computer Science	Programming 2	46.6	Pass
023	Maths	Trigonometry 7	81	Merit
596	Science	Physics 1	50	Pass

- 10 Write the results returned by the following SQL script:

```
SELECT Student ID, Subject
FROM MARKS
WHERE Subject = "Programming 2"
```

- 11 Write the results returned by the following SQL script:

```
SELECT Student ID, Subject, Grade
FROM MARKS
WHERE Grade = "Pass"
```

- 12 Complete the SQL script to display the Test name, percentage and grade where the percentage is more than 50.

```
..... Test name, Percentage, .....
FROM .....
WHERE Percentage .....
```

- 13 Complete the SQL script to display the student ID, subject and percentage where the subject is Science.

```
SELECT Student ID, ....., .....
..... MARKS
..... SUBJECT = .....
```

- 14 Complete the SQL script to display the student ID, test name and percentage where the percentage is less than or equal to 25.

```
....., Test name, Percentage
.....
WHERE ..... <= .....
```

Questions

These questions use the following database table, WEATHER, that store weather readings.

Date	Time	Temperature	Wind speed	Humidity
11/12/2019	10:00	20.3	22	59
11/12/2019	13:00	23.7	15	63
11/12/2019	15:00	24.0	13	61
11/12/2019	18:00	23.9	16	60
12/12/2019	15:00	18.4	20	52
12/12/2019	18:00	17.2	18	48

- 15** Write the results returned by the following SQL script:

```
SELECT Temperature, Wind speed, Date
FROM WEATHER
WHERE Date = 11/12/2019 AND time < 15:00
```

- 16** Write the results returned by the following SQL script:

```
SELECT Date, Time, Humidity
FROM WEATHER
WHERE Temperature > 20 AND Wind speed < 20
```

- 17** Write the results returned by the following SQL script:

```
SELECT Date, Time
FROM WEATHER
WHERE Temperature > 20 OR Wind speed >= 20
```



- 18 Complete the SQL script to display the time when the date is 12/12/2019 and the humidity is more than 50.

SELECT

FROM

WHERE= 12/12/2019 Humidity > 50

- 19 Complete the SQL script to wind speed and temperature, when the wind speed is either below 15 or above 20.

..... Wind speed,

FROM WEATHER

WHERE Wind speed 15 Wind speed 20



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11/12/2019	18:00	23.9	16	60
12/12/2019	15:00	18.4	20	52
12/12/2019	18:00	17.2	18	48

- 20** Write the results returned by the following SQL script:

```
SELECT Temperature, Wind speed, Date
FROM WEATHER
ORDER BY Temperature
```

- 21** Write the results returned by the following SQL script:

```
SELECT Date, Time, Temperature
FROM WEATHER
ORDER BY Humidity DESC
```

- 22** Complete the SQL script to display the date, wind speed and humidity in ascending order of wind speed.

```
..... Date, Wind speed, Humidity
..... WEATHER
..... Wind speed
```

- 23** Complete the SQL script to display the date, time and temperature, in descending order of humidity.

```
SELECT Date, Time, .....
FROM WEATHER
ORDER BY ..... ..
```

- 24** Complete the SQL script to display the temperature, wind speed and humidity in descending order of temperature.

```
....., Wind speed, .....
FROM .....
..... ..
```

Questions

These questions use the following database table, `ORDERS`, that stores details about orders from a shop.

Order ID	FirstName	LastName	NumberItems	TotalCost
15	Keanu	Kawai	5	20.59
16	Keanu	Kawai	10	101.12
17	Eka	Loke	3	30.50
18	Keanu	Kawai	2	14.00
19	Ariana	Wang	4	21.10
20	James	Smith	6	44.60
21	James	Smith	20	156.90
22	Ajay	Gupta	1	10.00

- 25 Write the results returned by the following SQL script:

```
SELECT SUM(NumberItems)
FROM ORDERS
```

- 26 Write the results returned by the following SQL script:

```
SELECT COUNT(NumberItems)
FROM ORDERS
```

- 27 Write the results returned by the following SQL script:

```
SELECT COUNT(NumberItems)
FROM ORDERS
WHERE NumberItems > 10
```

- 28 Write the results returned by the following SQL script:

```
SELECT SUM(TotalCost)
FROM ORDERS
WHERE FirstName = "Keanu"
```

- 29 Complete the SQL script to display how many orders James Smith has made.

```
SELECT ..... (FirstName)
FROM ORDERS
WHERE FirstName = "James" ..... LastName =
"Smith"
```

- 30 Complete the SQL script to calculate the total cost of orders where 5 or more items have been ordered.

```
SELECT ..... (TotalCost)
FROM ORDERS
..... NumberItems ..... 5
```

- 31 Complete the SQL script to calculate how many orders cost more than 50.00

```
SELECT ..... (NumberItems)
FROM .....
WHERE .....
```

EXAM-STYLE QUESTIONS

- 1 **Define** the terms record and field. Give an example of each. [2]
- 2 **Describe** the following data types and give an example for each.
 - Boolean
 - Real
 - Date / time
 - Text [4]
- 3 A cinema stores data about the films it is showing. It stores the screen number (for example, 1), the film (for example, The Dark), the date and time of the showing, the number of seats sold (e.g. 150).
Define the single-table database for the cinema by identifying the fields, data types, and primary key for the table. [7]

COMMAND WORDS

define: give precise meaning.

describe: state the points of a topic / give characteristics and main features.

CONTINUED

- 4 A hairdressers uses a database, APPOINTMENTS, to store upcoming appointments. Part of this database is shown:

Appointment ID	Date	Time	Employee ID	Customer first name	Customer last name	Appointment type
12	1/2/2020	14:00	JP	James	Smith	Cut
13	1/2/2020	14:30	AD	Ajay	Gupta	Cut and dry
14	1/2/2020	16:00	AD	Eka	Loke	Colour
15	1/2/2020	9:45	JP	Keanu	Kawai	Cut and dry
16	2/2/2020	11:15	JP	Ariana	Wang	Cut
17	2/2/2020	12:30	AD	Osandi	Bandara	Colour

- a **Identify** how many records are in the table. [1]
 b Identify how many fields are in the table. [1]
 c **State** what is meant by a primary key, *and* identify the primary key in the table APPOINTMENTS. [2]
 d Identify the output from this SQL script:

```
SELECT EmployeeID, Appointment type
FROM APPOINTMENTS
```

[2]

- e Identify the output from this SQL script:

```
SELECT COUNT(AppointmentID)
FROM APPOINTMENTS
WHERE Appointment type = "Cut"
```

[1]

- f Complete the SQL script to return the first and last name of all customers with appointments on 1/2/2020.

```
..... Customer first name, Customer last name
FROM APPOINTMENTS
WHERE .....
```

[4]

- g Complete the SQL script to return the number of appointments JP has on 1/2/2020.

```
SELECT ..... (Employee ID)
..... APPOINTMENTS
..... Employee ID .....
```

[5]

[Total: 16]

COMMAND WORDS

identify: name /
select / recognise.

state: express in
clear terms.

CONTINUED

- 5 A single-table database, SALES, stores details of people who have bought books online.

Sales ID	Book ID	Date	Customer ID	Quantity bought	Total Cost
5	10	3/3/2020	1GHF2	1	6.99
6	151	3/3/2020	34FDD	2	10.00
7	230	3/3/2020	34FDD	1	12.00
8	230	3/3/2020	1GHF2	1	12.00
9	10	3/3/2020	P0001	2	13.98
10	88	4/4/2020	15DTB	2	10.50
11	209	4/4/2020	1GHF2	3	12.00
12	151	4/4/2020	34FDD	5	25.00
13	28	4/4/2020	34FDD	2	20.00

- a Identify the most appropriate data types for the following fields:

- Sales ID
- Date
- Customer ID
- Quantity bought
- Total Cost

[5]

- b Give the result from the following SQL script:

```
SELECT Date, CustomerID
FROM SALES
WHERE CustomerID = "1GHF2"
```

[2]

- c Complete the following SQL script to return the customer ID and total cost of each order where more than 1 book were ordered.

```
SELECT Customer ID, .....
FROM .....
WHERE Total Cost .....
```

[4]

- d Give the result from the following SQL script.

```
SELECT Sales ID, BookID
FROM SALES
WHERE Date = 4 / 4/ 2020 AND Total Cost <= 15
```

[2]

COMMAND WORD

give: produce an answer from a given source or recall / memory.

CONTINUED

- e Complete the following SQL script to calculate the total number of books sold on 4/4/2020.

```
SELECT ..... (Quantity bought)
..... SALES
WHERE ..... = .....
```

[4]

- f Complete the following SQL script to calculate the total cost of all the orders.

```
SELECT ..... (Total Cost)
.....
```

[3]

[Total: 20]